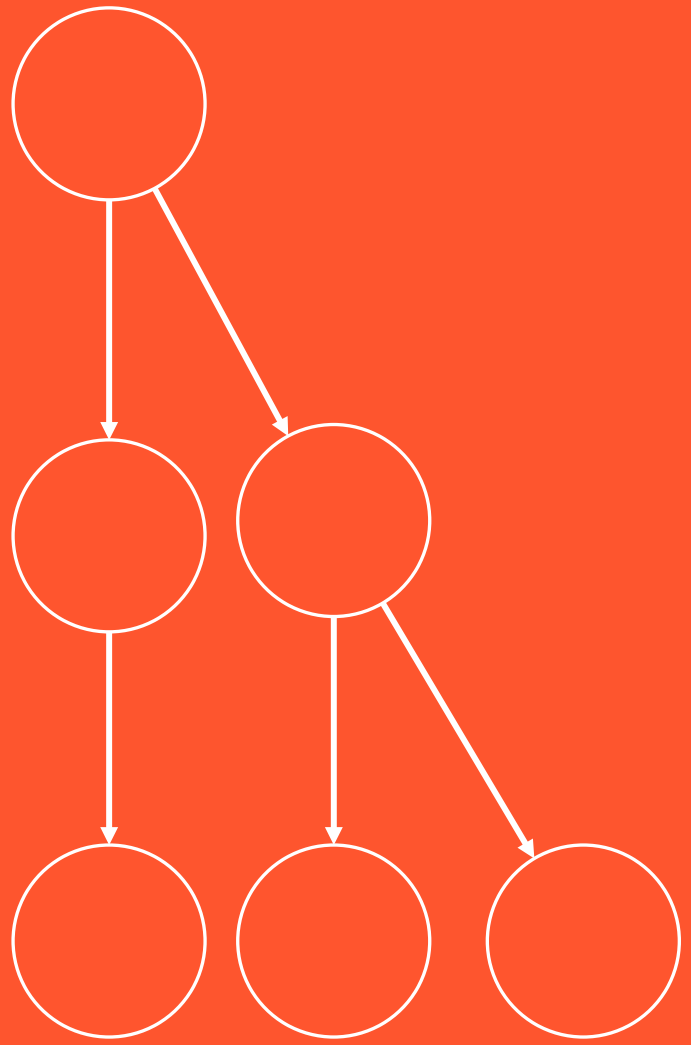


BCOG 100

Perception and Action

Lecture 8.1 — The Problems That Senses Solve

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

To Start: One Wave at Each Ear

In your group, three minutes.

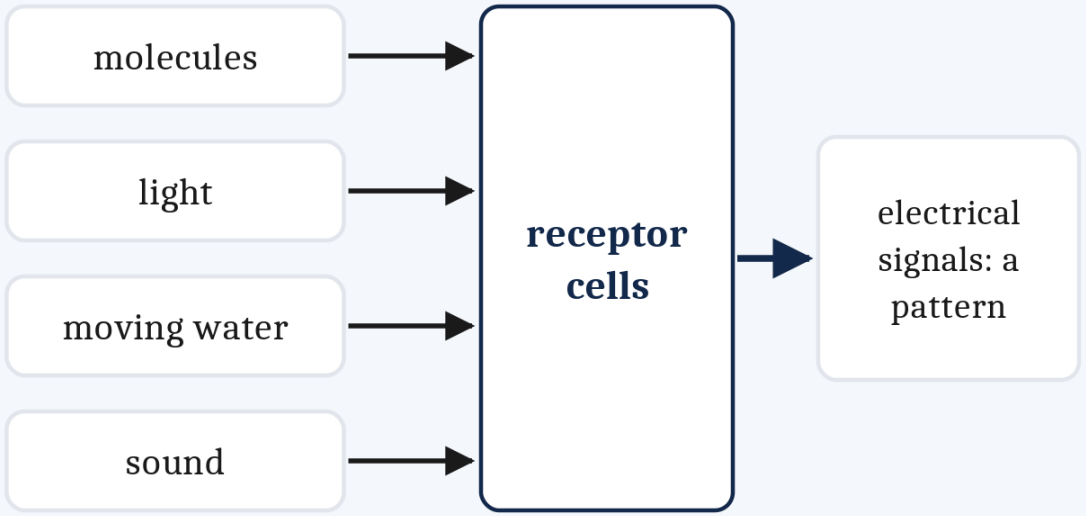
Picture a busy kitchen, eyes closed. A pan drops, someone is talking, and water is running. Each ear receives one wave, the sum of all of it.

How can anyone tell there are three things, and what each one is? What can the ears not tell, that the eyes could?

What We Set Aside in Module 7

Receptors have to make a pattern before a brain can use one. And a moving animal causes much of what reaches its own senses.

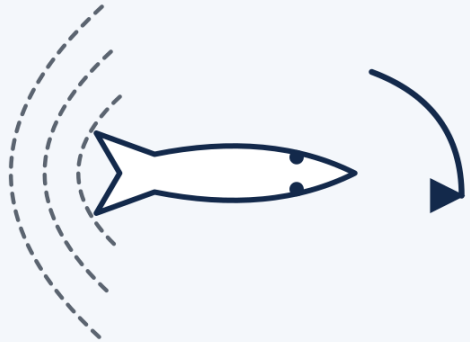
Before there is a pattern



In Module 7, we took the pattern as given. Each kind of signal tells the fish some things, and not others.

A fish that moves

Each tail beat stirs the water along its own lateral line.



Each turn sweeps the scene across its eyes.

By the time news of the eel arrives, the eel has moved on.



Much of what reaches a moving animal's senses was caused by the animal itself.

What Each Signal Carries

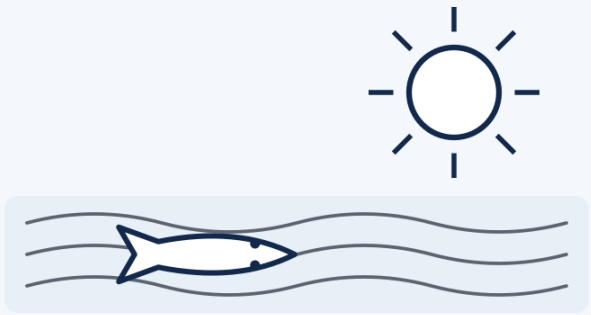
None of a sense's information is direct. Each signal carries some facts and not others, because of how it travels.

Signal	How it travels	Carries well	Leaves out
Light	Fast, in straight lines	The layout of surfaces at a distance, and precise direction	Anything behind an obstacle; everything, in the dark
Sound	Around obstacles, in the dark, from every direction at once	That something is happening, even out of sight, and what made it	Direction is not laid out; it has to be computed from the two ears
Chemicals	Slowly, drifting with the current, lingering	What something is	Where it is, except poorly
Touch and the lateral line	Contact with the body, or moving water within a body length or two	What is touching, and movement close by	Anything farther away

What an Eye Must Deliver Depends on Its Owner

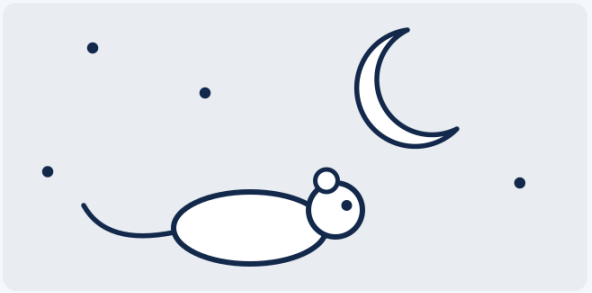
What a sense has to provide depends on how its owner lives.

A fish in shallow water



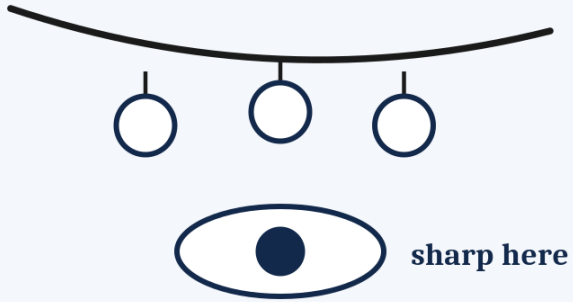
Sees ultraviolet light, which the human eye filters out before it reaches the receptors.

An early mammal



Probably small and active mostly at night, when sensitivity to dim light matters far more than color.

A primate in daylight



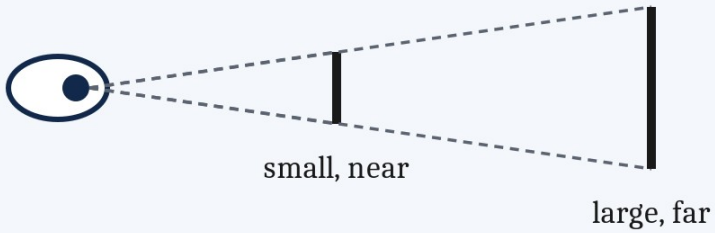
Picking fruit and insects, it needs sharp detail at the center of what it is looking at.

What a sense has to provide depends on how its owner lives, and vision is the clearest case.

Never Enough Information

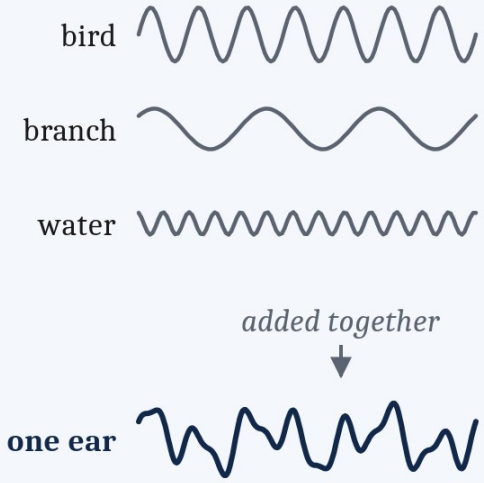
An **ill-posed problem**: the information available does not settle a single answer. Every sense faces a version of it.

Vision



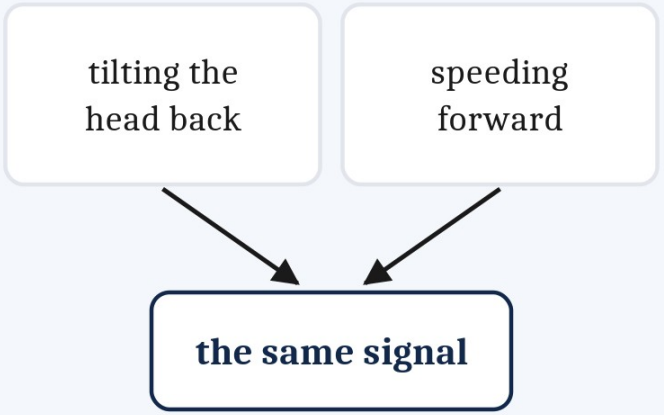
One flat image fits many different scenes.

Hearing



Many combinations of sources give the same wave.

Balance



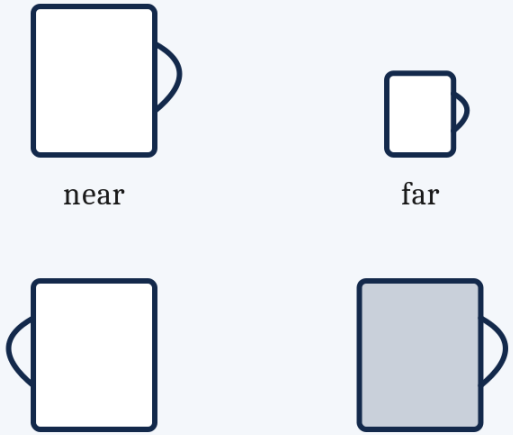
To a sensor that responds to acceleration, the two are the same.

What brains settle for:	which of two points is nearer, and an object's shape:	**reliable**	*"robustness over precision"*
	exact distance, even in a well-lit room:	**not reliable**	

The Same Thing, Again

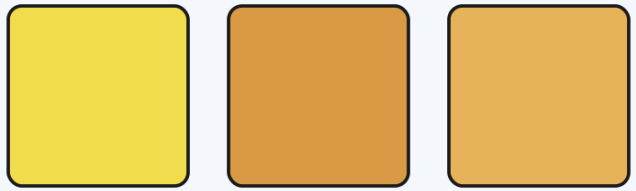
Recognizing the same thing across inputs that change completely is a problem every sense faces, in its own version.

One object, many images



Daylight can be up to a billion times brighter than starlight.

One color, many lights

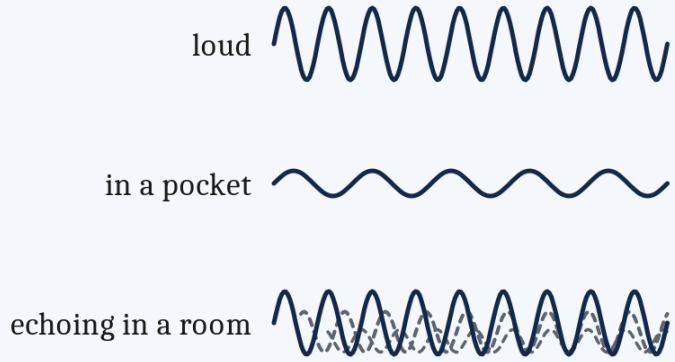


light reaching the eye from the banana

still looks yellow

The light mixes surface and lamp; nothing in it says how much is each.

One sound, many waves



the same ringtone

recognized at once, every time

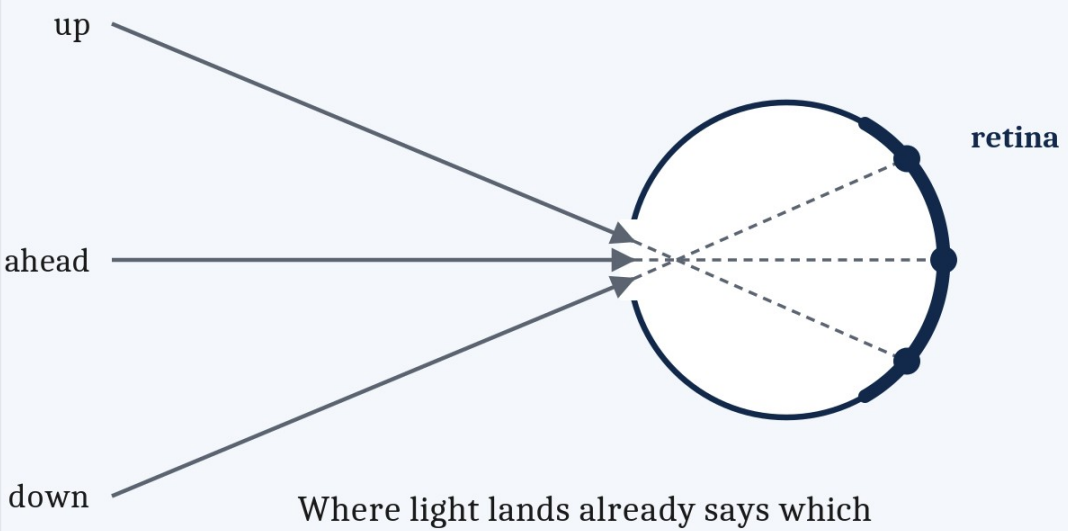
Each sense has its own version of the hardest generalization from Module 7:

the same thing, across inputs that change completely.

Where Things Are: Direction Comes Free

An image gives direction with no further computation. Distance is not on the retina, so it has to be computed, from differences between views.

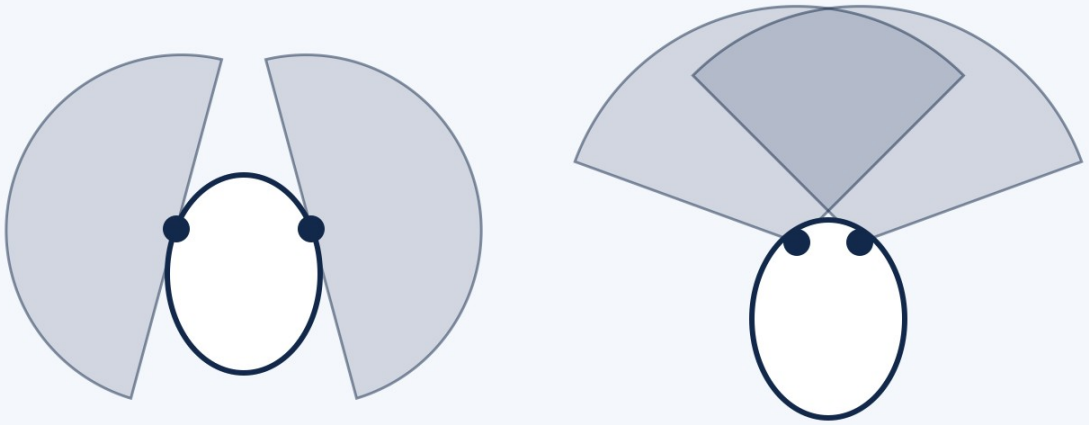
Direction comes free with an image



Where light lands already says which direction it came from.

Distance is not laid out on the retina.

Where the eyes sit



on the sides

a wide view, good for watching for predators

facing forward

overlapping fields (darker): most of the scene seen twice

Distance has to be computed, from differences between views: two eyes, or one eye that moves.

Hearing Has to Compute Even Direction

The **cochlea**, the mammalian organ of hearing, lays out its receptors by pitch, not by direction. Direction lies in differences between the two ears.

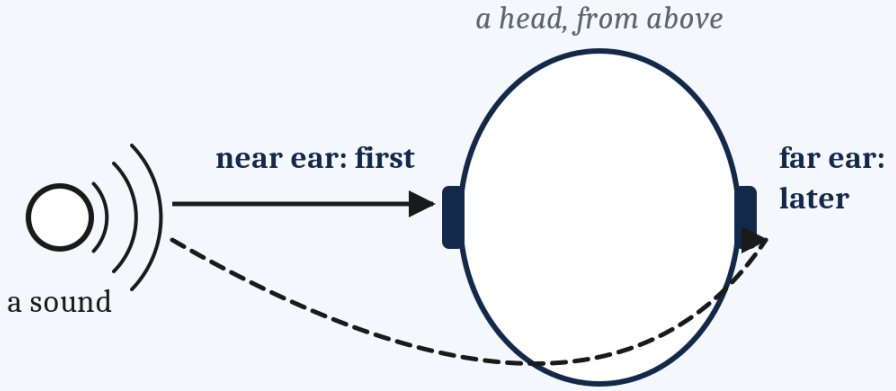
The cochlea's layout is by pitch



Which receptors respond says which frequencies are in a sound, not where the sound came from.

(the cochlea drawn uncoiled)

Direction lies in the difference



Straight to one side: about 0.7 ms later. A one-degree shift: about ten millionths of a second.

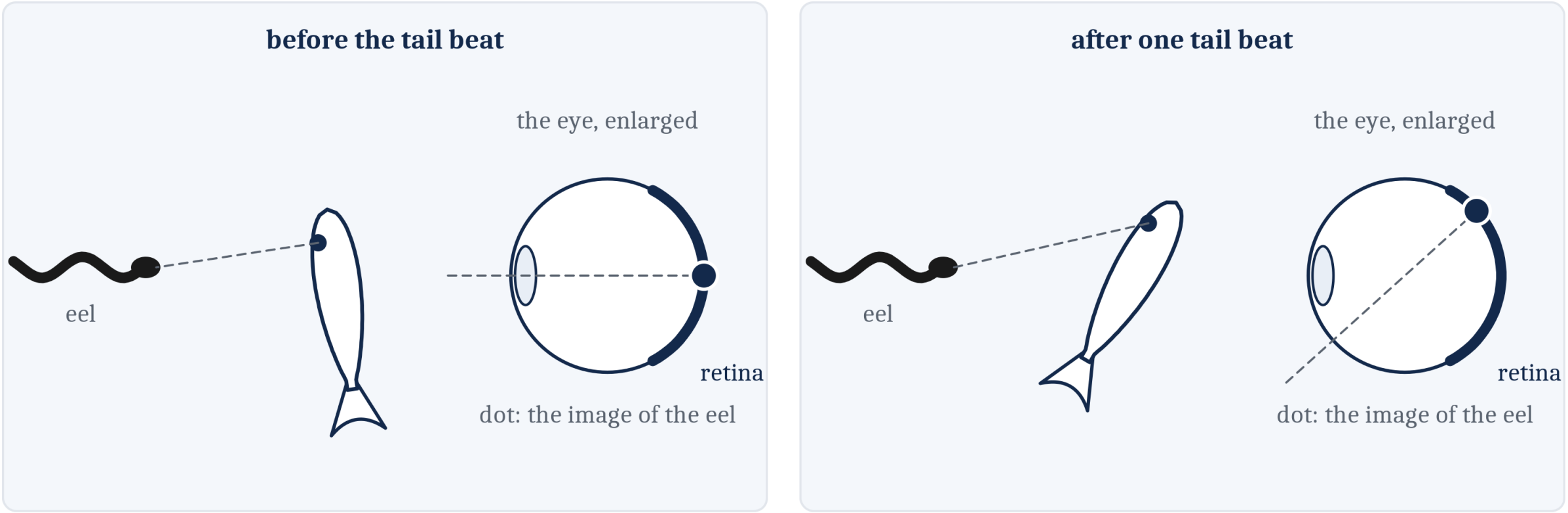
Mammals' ears are isolated from each other, so left and right exist only in the difference.

A lizard's eardrums are joined through its mouth, so each ear is directional by itself.

Direction on the Eye Is Not Direction in the World

Where an image falls gives its direction relative to the eye. Every movement of eye, head or body moves the image.

Where an image falls depends on where the eye points

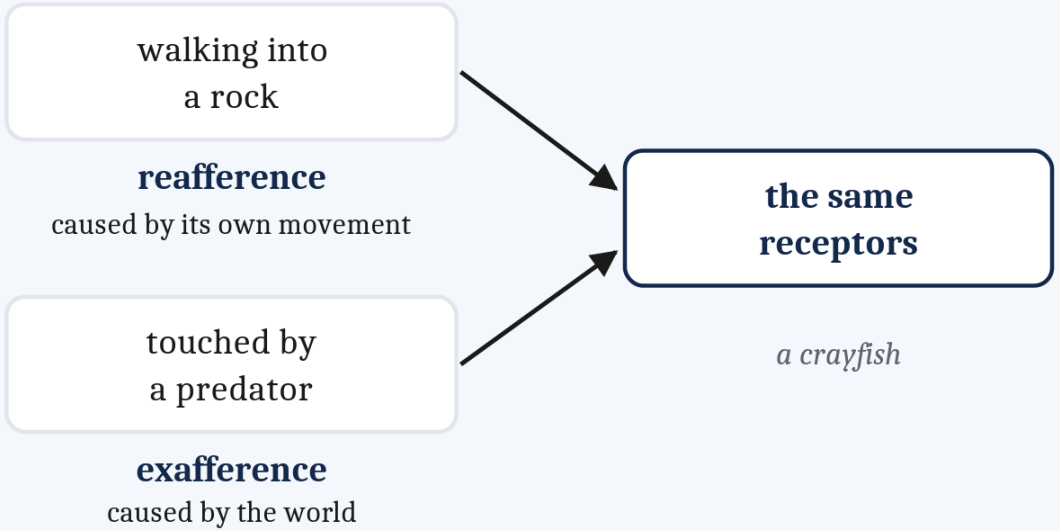


same eel, same place, different place on the retina

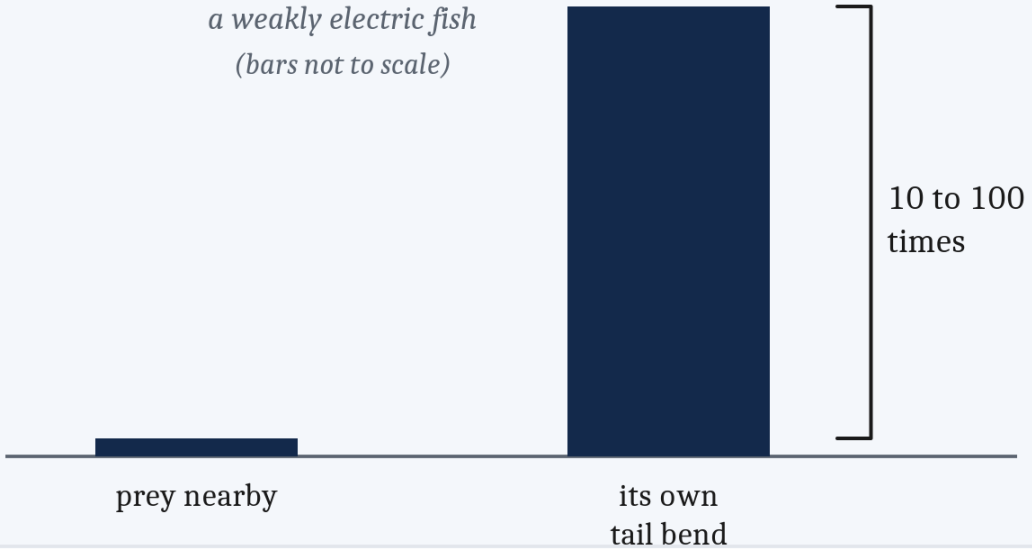
Signals of One’s Own Making

Reafference: sensation caused by the animal’s own movement. **Exafference**: sensation caused by the world, independent of what the animal does.

Same receptors, two causes



Self-caused signals can swamp the world's



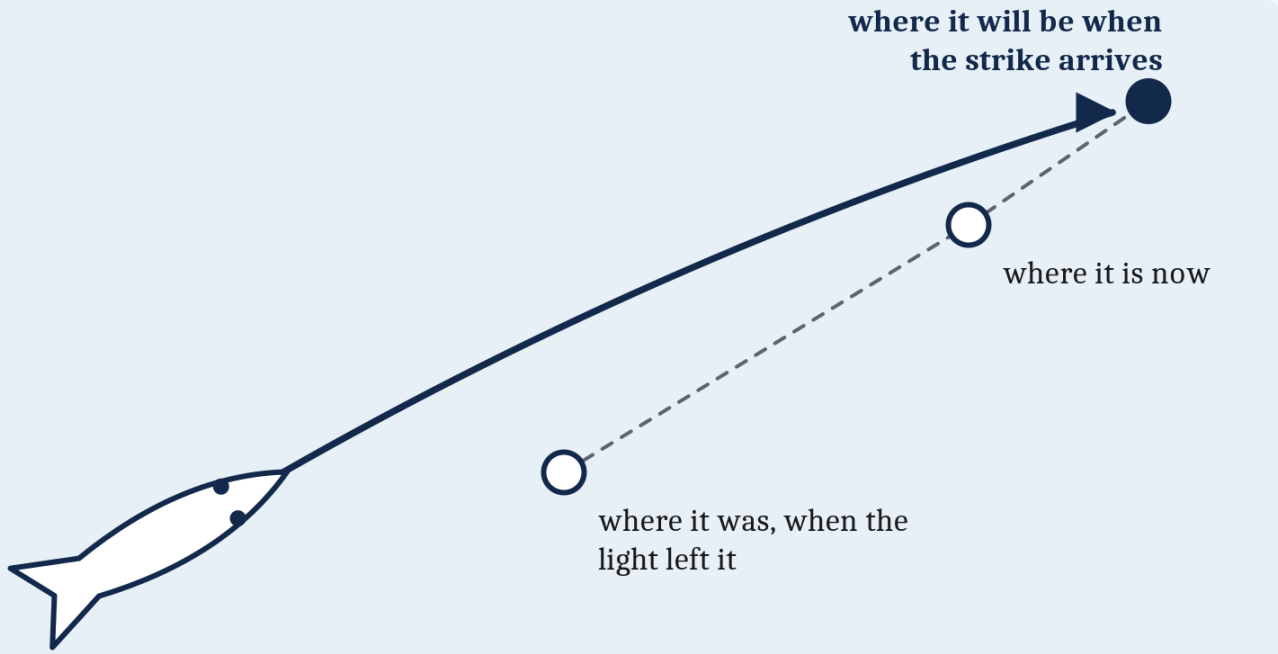
Two sources of information about its own movement

- the commands it sends to its muscles
- senses of the movement itself: vestibular system, proprioception

A Slow Body

A body acts on old information, keeps changing, and gets its signals in the wrong coordinates. A **saccade**, a quick jump of the eyes, is over before feedback could arrive.

The strike at a drifting water flea



Aim at where the flea will be, from signals about where it was.

Delay

An image slip takes about a tenth of a second to become a correction. Late feedback overshoots and oscillates.

Too fast for feedback

A saccade, a jump of the eyes, lasts tens of milliseconds.

A changing body

Growth, fatigue, injury, glasses: command-to-outcome must be relearned.

The wrong coordinates

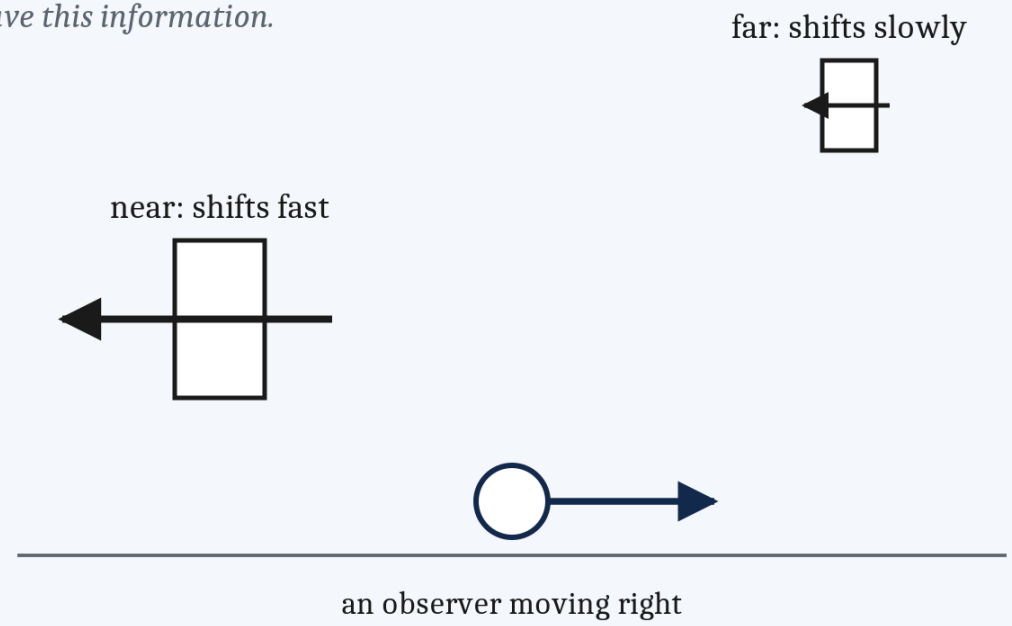
A place on the retina is not a set of joint angles.

Moving in Order to Perceive

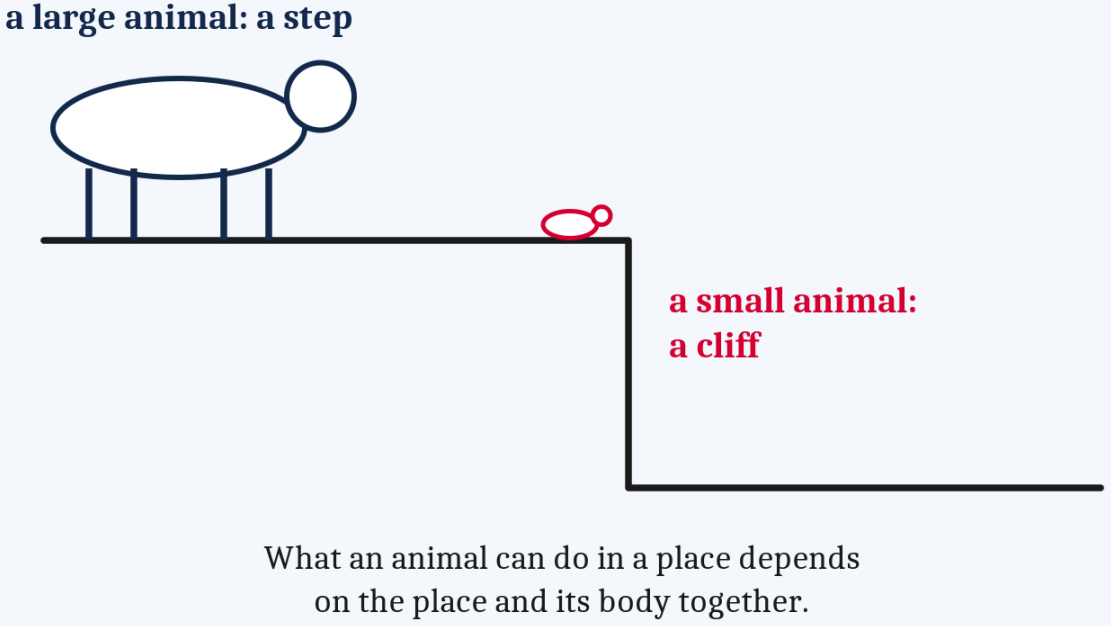
Movement also makes information. An **affordance** is an opportunity for action that a place or object offers to a particular body.

Moving makes information

A stationary animal does not have this information.



One drop, two bodies



What an animal can do in a place depends on the place and its body together.

Affordance: an opportunity for action that a place or object offers to a particular body.

A Camera That Holds Still

In your group, four minutes.

Many phone cameras steady a video when the hand holding them shakes, but not when a dog runs across the picture.

How could the phone tell the two apart? Which of the fish's two sources of information about its own movement does the phone have, and which does it lack?

And what about a slow, deliberate pan across a room?

Image Sources (1 of 2)

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3	Two things set aside in Module 7	fig-set-aside.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
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6	Three senses, three ill-posed problems	fig-ill-posed.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
7	Each sense's own invariance problem	fig-same-thing.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
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9	Hearing computes direction	fig-ears-where.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
10	Where an image falls depends on where the eye points	fig-where-on-the-eye.png	CC BY 4.0	Jon Willits — drawn for the Module 8 reading · own work
11	Self or world	fig-self-or-world.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work

Image Sources (2 of 2)

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12	The strike at a water flea	fig-slow-body.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	Moving makes information; one drop, two bodies	fig-step-or-cliff.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work